

EC101 Notes — Basic Concepts of Economics

Based on Chapters 2–4: Basic Problems of Economic Organization; Markets and Command; Elements of Supply and Demand

Chapter 2: Basic Problems of Economic Organization

A. The Three Fundamental Economic Problems

Every economic system — no matter how rich, poor, capitalist, or communist — must answer **three central questions**, because resources are scarce relative to wants:

1. **WHAT** — What kinds and quantities of goods and services shall be produced?
2. **HOW** — How shall resources (land, labor, capital) be used/combined to produce these goods?
3. **FOR WHOM** — For whom are the goods produced? (i.e., how is output/consumption distributed among individuals?)

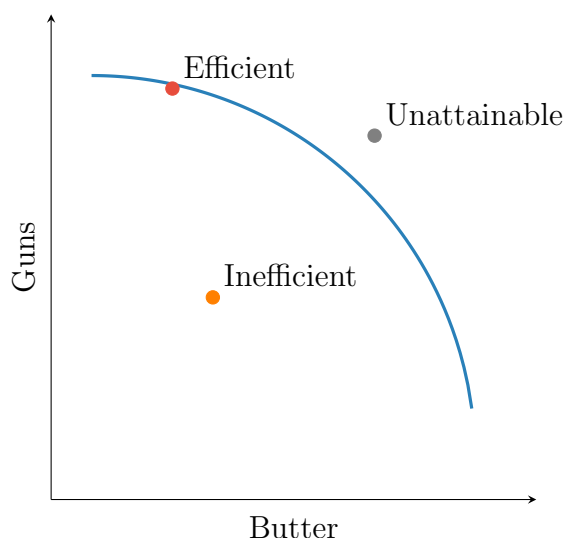
*These three questions exist in every society — a family, a firm, a college, an army, a command economy, or a market economy. What differs across systems is **who** answers them and **how**.*

Key Concepts

Term	Meaning
Scarcity	Resources (labor, land, capital) are limited relative to human wants — the root cause of all economic problems.
Free goods	Goods available in such abundance that no scarcity exists (e.g., air) — not subject to the economic problem.
Efficiency	Getting the maximum output from available inputs/resources — no waste.
PPF	A curve showing the maximum combinations of two goods an economy can produce with fixed resources and technology, assuming full and efficient use of resources.

Understanding the PPF

- **On the frontier** → resources are being used fully and efficiently.
- **Inside the frontier** → inefficiency or unemployment of resources.
- **Outside the frontier** → **not attainable** with current resources/technology.
- The PPF is generally **bowed outward (concave to origin)** due to the **law of increasing opportunity costs** — as you produce more of one good, you must give up increasing amounts of the other.



Shifts of the PPF & Opportunity Cost

- **Economic growth** shifts the *entire* PPF outward.
- A technological improvement in **only one good** rotates the frontier outward only along that good's axis.
- **Resource depletion** shifts the PPF inward over time.
- **Opportunity Cost:** The value of the next-best alternative given up. Example: A compulsory (low-pay) army hides the true opportunity cost (foregone productive alternatives) because the budget figure looks small.

Chapter 3: Markets and Command in a Modern Economy

A. Definitions & The Market Mechanism

- **Market mechanism:** Individual consumers and businesses interact through markets to solve the central problems — **no central authority directs it.**
- **Command economy:** Resource allocation is determined by government directives.
- **Mixed economy:** Private markets operate via the "invisible hand," while government shapes outcomes through regulation and taxation.

How the Market Solves WHAT, HOW, and FOR WHOM:

1. **WHAT is produced** → Determined by **consumers' "dollar votes."** Firms chase profit.
2. **HOW goods are produced** → Determined by **competition.** Firms must minimize costs using the most efficient production method. Prices act as signals (e.g., land-intensive vs. labor-intensive farming).
3. **FOR WHOM goods are produced** → Determined by **supply and demand in factor markets.** $\text{Income} = (\text{quantity of factors owned}) \times (\text{price of those factors}).$

Perfect vs. Imperfect Competition

- **Perfect competition:** So many firms/buyers that **no single one can affect the market price.**
- **Imperfect competition:** A firm **can** influence price (e.g., airlines, labor unions).
- **Monopoly:** A single supplier controls the price entirely. Leads to higher prices and lower output. Government responds with antitrust laws.

B. The Economic Role of Government — Market Failures

1. **Imperfect Competition / Monopoly:** Government uses antitrust and regulation.
2. **Externalities:** Uncompensated costs/benefits to third parties. (e.g., Pollution = negative externality). Market over-produces negative externalities.
3. **Public Goods:** Non-excludable/non-rival goods (like national defense) which the market under-provides.
4. **Inequity in Income Distribution:** Tools include progressive taxation, transfer payments (safety net), and in-kind transfers.

5. **Macroeconomic Instability:** Government manages inflation/unemployment via monetary and fiscal policy.

“To operate a modern economy without both [market and government] is like trying to clap with one hand.”

Chapter 4: The Elements of Supply and Demand

The Market Mechanism operates via prices, sales, and profits acting as **signals** that automatically reallocate resources.

The Demand Curve

Shows the relationship between **price** and **quantity demanded**.

- **Law of downward-sloping demand:** As price falls, quantity demanded rises.
- **Substitution effect:** Consumers switch to cheaper substitutes.
- **Income effect:** A price rise reduces real purchasing power.

Factors shifting Demand: Income, Population size, Price of related goods (substitutes/complements), Tastes, Special factors (e.g., weather).

*Note: A change in a good's own price causes movement **along** the curve. A change in other factors **shifts** the curve.*

The Supply Curve

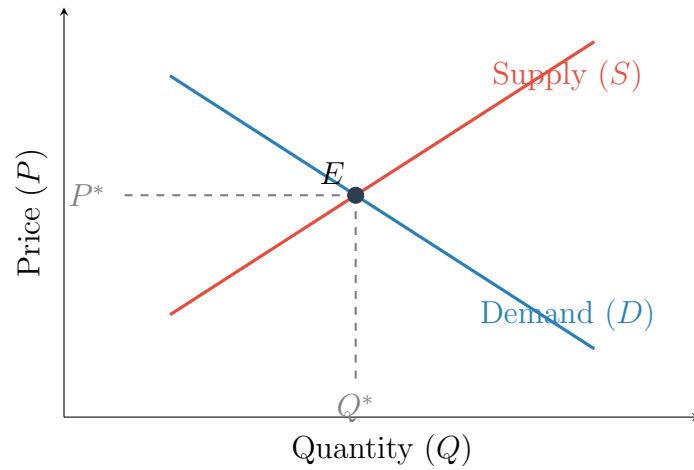
Shows the relationship between a good's price and the quantity producers are willing to supply. Slopes upward due to the **law of diminishing returns**.

Factors shifting Supply: Cost of production, Technology, Input prices, Foreign competition.

Market Equilibrium & Graphical Representation

Equilibrium occurs where Quantity Demanded (Q_d) = Quantity Supplied (Q_s).

- **Surplus:** $Q_s > Q_d$ at a given price $\rightarrow$ price falls.
- **Shortage:** $Q_d > Q_s$ at a given price $\rightarrow$ price rises.



Common Pitfalls in S&D Analysis

- **Movement ALONG vs. SHIFT:** “Increase in demand” = curve shifts right. “Increase in quantity demanded” = movement along a fixed curve (usually due to a price change caused by a supply shift).

Important Subjective Questions

1. **Explain the concept of Opportunity Cost using a real-world example other than the military.**

Answer approach: Define it as the value of the next best alternative foregone. Example: The opportunity cost of attending college is not just tuition and books, but also the foregone wages a student could have earned working full-time.

2. **Differentiate between a shift in the demand curve and a movement along the demand curve. What causes each?**

Answer approach: A movement along the curve is caused *only* by a change in the good's own price (*ceteris paribus*). A shift of the entire curve is caused by changes in non-price determinants (income, tastes, prices of related goods, etc.).

3. **Why does the Production Possibility Frontier (PPF) bow outward?**

Answer approach: Due to the law of increasing opportunity cost. Resources are not perfectly adaptable to producing both goods. As you produce more of Good A, you must give up resources that were better suited for Good B, resulting in larger sacrifices of Good B for each additional unit of A.

4. **How does the market mechanism solve the "FOR WHOM" problem, and what are its limitations?**

Answer approach: It solves it through factor markets determining wages, rents, and profits, which dictate purchasing power. Limitation: it does not guarantee equity; those without scarce skills or capital may receive very little income, requiring government intervention.

Multiple Choice Questions (MCQs)

1. Which of the following would cause a *movement along* the supply curve for coffee?

- A A technological improvement in coffee harvesting.
- B A change in the price of coffee.
- C An increase in the wage paid to coffee bean pickers.
- D A drought destroying half the coffee crop.

2. The production possibility frontier shifts outward when:

- A Unemployment increases.
- B Resources are depleted.
- C There is technological advancement.
- D The country produces more of one good and less of another.

3. If the government imposes a strict environmental regulation on a factory, this is an attempt to correct:

- A A public good.
 - B A negative externality.
 - C A positive externality.
 - D Imperfect competition.
4. A bad harvest shifts the supply of wheat to the left. According to the "common pitfall" logic, what happens next?
- A The demand curve shifts left, lowering the price back to normal.
 - B The demand curve shifts right, raising the price further.
 - C The supply curve shifts back right due to higher prices.
 - D The equilibrium price rises, causing a movement along the demand curve.
5. Which of the following is an example of **capital** in economics?
- A A farmer's labor.
 - B A naturally flowing river used for irrigation.
 - C A tractor purchased to plow fields.
 - D Money in a checking account.
6. In a perfectly competitive market, an individual seller:
- A Can influence the market price by changing output.
 - B Must accept the market price as given.
 - C Produces a unique product.
 - D Earns monopoly profits.

MCQ Answers

- 1. **B** (A change in the good's own price causes movement along the curve; the others shift the curve).
- 2. **C** (Technology and more resources cause economic growth, shifting the PPF outward).
- 3. **B** (Pollution is a classic negative externality; regulations force the factory to internalize the cost).
- 4. **D** (A leftward supply shift raises the price, which reduces the *quantity demanded* via a movement along the demand curve, not a shift of demand itself).
- 5. **C** (Capital is a produced factor of production used to make other goods. A river is land, labor is human effort, and money is a medium of exchange, not capital itself).
- 6. **B** (In perfect competition, individual buyers and sellers are price takers).